1.

```
#include<bits/stdc++.h>
using namespace std;
template< class T>
class node{
private:
    T data;
    node<T>* next;
/*     node(int _data):data(_data),next(0){}
    node():data(0),next(0){}  */
    template <class TT> friend class Queue;
};
template < class T>
class Queue{
public:
    Queue():front(NULL),rear(NULL),size(0){};
    void enqueue(T p){
    node<T>* ptr = new node<T>[1];
    ptr->data = p;
    if(IsEmpty()){
        front = ptr;
        rear = ptr;
    }else{
        rear->next = ptr;
        rear = rear->next;
        size++;
    }
}
    void dequeue(){
    if(IsEmpty()){
        cout << "queue is empty";
    }else{
        node<T> *deleteNode = front;
        front = front->next;
        delete deleteNode;
        deleteNode = NULL;
        size--;
    }
```

```
}
    bool IsEmpty(){
    if(front == NULL && rear == NULL){
        return true;
    }else return false;
}
    void top(){
    if(IsEmpty()){
        cout << "queue is empty";
    }else{
        cout << front->data;
    }
}
private:
    node<T>* front;
    node<T>* rear;
    int size;
};
int main(){
    Queue<int> q;
    q.enqueue(5);
    q.enqueue(2);
    q.enqueue(13);
    q.top();
    q.dequeue();
    cout << endl;
    q.top();
}
```

2.

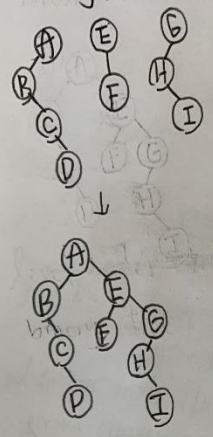
```
#include<bits/stdc++.h>
using namespace std;
struct node{
     int data;
     struct node* left;
     struct node* right;
};
struct node* newNode(int data){
```

```
    node* node = new struct node();
    node->data = data;
    node->left = NULL;
    node->right = NULL;
    return(node);
}
void printInorder(node* node){
    if (node == NULL) return;
    printInorder(node->left);
    cout << node->data << " ";
    printInorder(node->right);
}
void swapTree(node* node){
    if(node==NULL) return;
    struct node* temp=node->left;
    node->left=node->right;
    node->right=temp;
    swapTree(node->left);
    swapTree(node->right);
}
int main(){
    struct node *root  = newNode(1);
    root->left         = newNode(2);
    root->right        = newNode(3);
    root->left->left   = newNode(4);
    root->left->right  = newNode(5);
    cout << endl;
    cout << "Inorder of the binary tree "<< endl;
    printInorder(root);
    swapTree(root);
    cout << endl;
    cout << "After swap of the tree " << endl;
    printInorder(root);
    return 0;
}
```

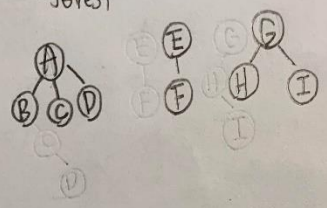
3.

3,

binery tree



forest



level order traversal

binery tree

ABECFGDHI

forest

AEGBCDFHI

From the above examaple we could know that the level-order traversal of forest and that of its corresponding binary tree do not necessarily yield the same result.

4.

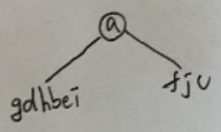
4.

Ex

inorder = gdhbeiafjc
preorder = abdgheicfj

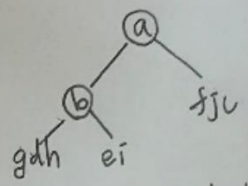
Scan the preorder left to right using the inorder to separate left and right subtrees.

a is the root of the tree; gdhbei are in the left subtree; fjc are in the right subtree.



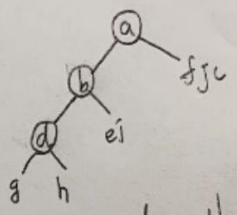
preorder = abdgheicfj

b is the next root; gdh are in the left subtree; ei are in the right subtree



preorder = abdgheicfj

d is the next root; g is in the left subtree; h is in the right subtree



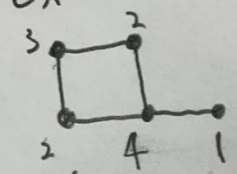
by keep doing the procedure

⋮

we could construct a unique binery tree.

5.

Ex



Sum of degree : $1+4+2+3+2=12$

$|E|=6$

⟨Pf⟩ In any graph G, $\sum_{v \in V(G)} \deg(v) = 2|E(G)|$

Let $e \in E(G)$ be any edge

Then e has 2 ends, say u and w (so $e=uw$)

when we sum the degrees of the vertices,

edge e gets counted twice (once in $\deg(u)$ term, once in $\deg(w)$ term)

Similary, every edge gets counted twice #

6.

6,

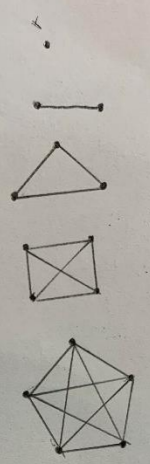
1 vertices — 0 edge

2 vertices — 1 edge

3 vertices — 3 edge

4 vertices — 6 edge

5 vertices — 10 edge



n,

⟨Pf⟩ Complete graph means that every vertex is connected with every other vertex.

If taking one vertex of the graph, we will get $(n-1)$ outgoing edges from that vertex.

now having n vertices in total, there might be $n(n-1)$ edges in total, but we counts every edge twice, since every edge going out from one vertex is an edge going into another vertex. Hence dividing the result by 2.

$\therefore$ we would get $\frac{n(n-1)}{2}$ #

+2=12

7.

```
#include<iostream>
#include<list>
using namespace std;

class graph{
private:
     int V;
     list<int> *adj;
public:
     graph(int V);
     void insertEdge(int v,int x);
     void BFS(int s);
};

graph::graph(int V){
     this->V = V;
     adj = new list<int>[V];
}

void graph::insertEdge(int v,int x){
     adj[v].push_back(x);
}

void graph::BFS(int s){
     bool *visit = new bool[V];
     for(int i=0;i<V;i++){
          visit[i] = false;
     }
     list<int> queue;
     visit[s] = true;
     queue.push_back(s);
     list<int>::iterator i;
     while(!queue.empty()){
          s = queue.front();
          cout << s << " ";
          queue.pop_front();
```

```
        for(i = adj[s].begin();i!=adj[s].end();++i){
            if(!visit[*i]){
                visit[*i] = true;
                queue.push_back(*i);
            }
        }
    }
}
int main(){
    graph g(6);
    g.insertEdge(0,1);
    g.insertEdge(0,2);
    g.insertEdge(1,2);
    g.insertEdge(1,3);
    g.insertEdge(2,3);
    g.insertEdge(3,4);
    g.insertEdge(4,0);
    g.insertEdge(4,1);
    g.insertEdge(4,5);

    g.BFS(1);
}
```

8.

8.

Total number of spanning tree in a graph

We could now that for n vertices, the total number of spanning trees is n^{n-2} where n is the number nodes in the graph

for $n=2$, the number of spanning trees is $2^0 = 1 = 2^{n-1}-1$ when $n=2$

for $n=3$

$3^{3-2} = 2^{3-1}-1$

for $n=4$

$4^{4-2} > 2^{4-1}-1$

for $n \geq 4$, we can conclude that

$n^{n-2} > 2^{n-1}-1$ (according to Mathematical Induction)

whereas when $n=2,3$

the number of spanning tree is at least

$2^{n-1}-1$ #

9.

```
#include<iostream>
#include<vector>
#include<list>
#include<queue>
#include<iomanip>
using namespace std;
class graph{
private:
    int V;
    vector<list<int>> adj;
    int *color;      //0:white, 1:gray, 2:black
    int *predecessor;
    int *discover;
    int *finish;
public:
```

```
    queue<int> q;
    graph():V(0){};
    graph(int n):V(n){
      adj.resize(V);
    };
    void insertEdge(int v,int x);
    void DFS(int s);
    void DFSVisit(int vertex,int &t);
};

void graph::insertEdge(int v,int x){
  adj[v].push_back(x);
}
void graph::DFS(int s){
  color = new int[V];
  discover = new int[V];
  finish = new int[V];
  predecessor = new int[V];

  int t = 0;
  for(int i=0;i<V;i++){
    color[i] = 0;
    discover[i] = 0;
    finish[i] = 0;
    predecessor[i] = -1;
  }
  int i = s;
  for(int j=0;j<V;j++){
    if(color[i]==0){
      DFSVisit(i,t);
    }
    i = j;
  }
}
void graph::DFSVisit(int vertex,int &t){
  color[vertex] = 1;
  discover[vertex] = ++t;
  cout << vertex << " ";
```

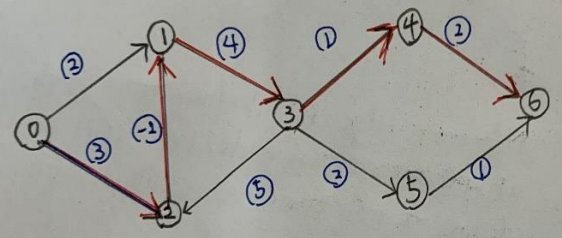
```
  for(list<int>::iterator itr =
adj[vertex].begin();itr!=adj[vertex].end();itr++){
    if(color[*itr]==0){
      predecessor[*itr]=vertex;
      DFSVisit(*itr,t);
     }
    }
  color[vertex] = 2;
  finish[vertex] = ++t;
}
int main(){
  graph g(6);
  g.insertEdge(0,1);g.insertEdge(0,2);g.insertEdge(0,3);
  g.insertEdge(1,4);g.insertEdge(2,4);g.insertEdge(2,5);
  g.insertEdge(3,4);g.insertEdge(3,5);
  g.DFS(0);
}
```

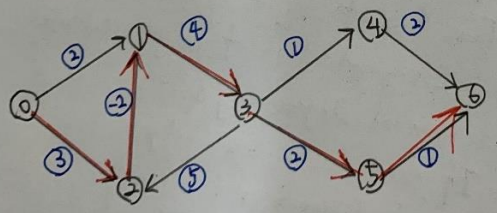
10.

10,

① Shortest path is a path that covers all the nodes with minimum weight values a cycle.
According to the shortest path algorithm, always move towards a smaller weight edge from starting vertex.
However, we can't find the shortest path if we follow the algorithm, since we must follow the longer weight edge and thus find the shortest path.

②





Shortest path

0→2→1→3→4→6 or 0→2→1→3→5→6

are both 8 which is shortest path